

Paper BIO-02: Adaptive Flux Limitation Across Living Domains: A CDFD Model of Flow, Constraint, Response, and Memory

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Abstract

This paper presents **Adaptive Flux Limitation (AFL)** as the Part III biology-and-medicine model inside Constraint-Driven Flux Dynamics (CDFD). CDFD supplies the flow-constraint-memory grammar: physiological flow Φ , constraint C , surface responsiveness S , and structural memory M_s . The Runtime citation anchors the CDFL implementation, while clinical interpretation remains separate from software output and bedside validation.

AFL-1 defined Adaptive Flux Limitation using coupled throughput Φ , constraint C , and ratio $\Psi_s = (\Phi/C) \cdot S \cdot M_s$. A remaining question is whether that **structure** is restricted to metabolism or recurs when (Φ, C) are reinterpreted across physiology.

We show—at the level of a shared dynamical skeleton plus domain-specific parameter maps—that metabolism, neural excitation, circulation, immunity, development, and ecology can all be read as AFL systems: the **same** update pattern for C coupled to transport of Φ , with interpretation changing by domain. An active-numbered public supplement should run four stylized domains with identical equations and distinct (α, β) and boundary drives.

Status: Inferred

Within the extended framework of this series, biological networks are modeled as *Adaptive Surfaces* that dynamically integrate physiological load and retain structural memory. The system's health is governed by the adaptive ratio $\Psi_s = (\Phi/C) \cdot S \cdot M_s$, where homeostasis ($\Psi_s \approx 1.0$) is maintained near the stable attractor ($S \cdot M_s \approx 1.0$), and disease emerges as surface dysregulation when maladaptive memory locks tissues into chronic constraint.

Greek-symbol convention. Unless a manuscript states a tighter equation-local meaning, Greek symbols are local CDFD variables or coefficients, not universal constants. Here Φ denotes driving flux, Ψ_s denotes the adaptive operating ratio, Ω denotes a domain or state space, and Λ denotes the Life Number where used. The coefficients α , β , and γ denote accumulation or reinforcement, relaxation or decay, and diffusion, coupling, or redistribution. The symbols δ and ϵ denote perturbation or tolerance scales, while η , λ , μ , ν , ρ , σ , τ , and ϕ denote local efficiency, rate, baseline, frequency, density or correlation, noise or variance, time-scale, and phase or field terms. Subscripts restrict any coefficient to the named pathway, tissue, domain, or experiment.

Clinical Reader’s Guide

This paper asks whether AFL is only a metabolism story or whether it is a wider biological pattern. The intended reader should track what remains constant and what changes across domains. What remains constant is the relation between flow, constraint, response, and memory. What changes is the measurable proxy: glucose disposal in metabolism, erythropoietic throughput in anaemia, immune effector activity in inflammation, growth signalling in cancer, membrane potential in bio-electric regulation, or ecological exchange in networks.

The clinical importance is classification. A conventional diagnosis tells the organ or disease family. AFL asks for the limiting axis. Two patients with different diagnoses may share a dominant constraint state; two patients with the same diagnosis may fail therapy for different reasons. The framework becomes useful only if it improves that stratification beyond standard disease labels.

The paper should be read as a hypothesis map. It does not say that all biology is the same. It says that many biological systems may reuse a common control problem: how to sustain useful throughput without damaging the medium that carries it.

1 Introduction

Adaptive Flux Limitation (AFL) is the named model used in this paper; CDFD is the parent framework that supplies the variables, closure logic, and claim discipline. The biological or medical question is posed first as AFL, then interpreted as a CDFD model of flow, constraint, surface responsiveness, and structural memory.

1.1 Formal Symbol Rigor: The Extended CDFD Paradigm

In strict alignment with the Fundamental Physics (Part I) and Origins of Life (Part II) archives, this biological translation formally preserves the exact Mujjabi transport tensor structure:

- Φ (**Flow Intensity**): The physiological or biochemical drive (e.g., nutrient flux, signaling rate).
- C (**Constraint / Pathway Resistance**): The biological resistance regulating the flow. It dynamically evolves via the standard closure: $dC/dt = \alpha\Phi - \beta C$.
- S (**Surface Responsiveness**): The active response capability of the biological medium.
- M_s (**Surface Memory**): Structural hysteresis or maladaptive drift (e.g., chronic scarring, epigenetic locking) with a finite recovery time τ_M .
- $\Psi_s = (\Phi/C) \cdot S \cdot M_s$ (**Operating State Ratio**): The dimensionless index of biological health. $\Psi_s \approx 1$ defines homeostasis ($\Psi_s \approx 1.0$); $\Psi_s \gg 1$ defines pathological overload.

Biology is partitioned into specialties with different vocabularies. Nevertheless, many systems show nonlinear saturation, bottlenecks, adaptive feedback, and stabilization bands. AFL-1 argued that throughput–constraint coupling is one parsimonious lens for those patterns in a single metabolic-style narrative.

Open gap addressed here: Do analogous assignments of (Φ, C) produce *isomorphic* update laws across domains? We answer “model-level yes” for the class of systems simulated here; empirical universality remains to be tested.

Status: Hypothesis

2 Model and Assumptions

2.1 Biological Translation of the Muijabi Capacity Law [Muijabi \[2026\]](#)

In Part I, the Muijabi Capacity Law [Muijabi \[2026\]](#) dictates that local transport becomes non-linear as the flux-to-constraint ratio approaches unity ($J/C \rightarrow 1$). In the biological realm, this explains metabolic saturation: as nutrient flow Φ scales up, the mitochondrial and enzymatic constraint C adapts upward, capping the throughput and triggering bottleneck responses (e.g., insulin resistance (chronic C_E upregulation)).

2.2 Shared transport and closure

Let φ denote a local driving potential appropriate to the domain (chemical, electrical, mechanical, or informational). A linear resistive closure is

$$\mathbf{J}_{\text{flux}} = -\frac{1}{C} \nabla \varphi. \quad (1)$$

Throughput intensity Φ denotes the domain-specific magnitude of transport (substrate flux, ion current magnitude, perfusion, signaling rate, trophic transfer). Constraints C aggregate resistance, saturation, suppression, or environmental limits. Legacy manuscripts used C for adaptive resistance; we continue to identify $C \equiv C$.

Status: Derived (definition)

2.3 Adaptive constraint dynamics

The AFL closure used in AFL-1 and in the supplement is

$$\frac{dC}{dt} = \alpha \Phi - \beta C, \quad (2)$$

with nonnegative effective (α, β) . Spatial discretizations replace Φ by lattice-redistributed fields; see script.

Status: Derived (given the adopted closure)

2.4 Assumptions B1–B3

- **B1** Each domain admits a reduction to effective (Φ, C) at the scale of interest.
- **B2** Equation (2) (or its spatial analogue) is an adequate first-order model for comparative numerics.
- **B3** Domain differences appear primarily in (α, β) , boundary drives, and units—not in the functional form of the closure.

Status: Open (B1–B3 can fail where control is strongly nonlocal or digital)

3 Deep Analysis: Cross-System CDFT/AFL Manifestations

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Figure 1: **CDFT/AFL Input–Output Dissociation Across Seven Classical Biological Systems.** Six panels show $J_i(I) = J_{\text{eff}} \cdot I / (K^{(i)} + I)$ for normal (solid) and constrained/disease (dashed) states; shaded gap = dissociation zone. In CDFT language: solid line = low C (high D_{eff}); dashed = elevated C (reduced D_{eff}); gap = the adaptive medium has become more resistive. Microbiological panel: resistance cycling ODE dynamics. Summary panel (bottom right): shared CDFT structure $F = \nabla X / C(\Omega, t)$. **KEY:** Dominant C component annotated in *italic* per panel.

3.1 Haematologic system: ESA hyporesponsiveness as C_M dominance

Erythropoiesis requires high methylation capacity (rapid DNA synthesis through one-carbon metabolism [Stover, 2009]), energetic capacity, and redox balance [Vaziri, 2004]:

$$J_{\text{RBC}} \leq \frac{\nabla X_{\text{erythro}}}{C_M + C_E + C_R} \quad (3)$$

In CKD, uraemia simultaneously increases C (through mitochondrial dysfunction and oxidative stress) and raises J_{req} (through inflammatory cytokine activity). ESA dose escalation increases ∇X (the hormonal driving gradient) without reducing any C component. The Dissociation Theorem directly predicts $dJ_{\text{RBC}}/dI \rightarrow 0$ at the haemoglobin plateau. A characteristic consequence is the serum–intracellular dissociation: normal circulating folate and B12 (normal ∇X_M) while intracellular C_M remains elevated because transport constraints prevent adequate one-carbon flux. Serum biomarkers measure ∇X ; AFL requires measurement of C .

3.2 Metabolic system: insulin resistance (chronic C_E upregulation) as protective C_E, C_R upregulation

Glucose utilization generates ATP but simultaneously produces ROS proportional to flux. When substrate flux approaches mitochondrial processing capacity, C_E and C_R rise protectively (fast dynamics: $dC/dt = \alpha F - \beta C$):

$$J_{\text{glucose}} \leq \frac{\nabla X_{\text{glucose}}}{C_E + C_R} \quad (4)$$

GLUT4 internalization, IRS-1 serine phosphorylation, and ceramide accumulation [Petersen and Shulman, 2018, Samuel and Shulman, 2016] are the molecular implementation of AFL fast dynamics — the cellular mechanism by which C is raised to protect structural integrity. They are not the origin of insulin resistance (chronic C_E upregulation) but its molecular execution. In CDFT language: the medium has raised C to reduce D_{eff} because the flux it was carrying was threatening structural damage. Chronic low-grade inflammation (system-wide constraint C amplification) simultaneously raises C through ROS production and inflammatory mediators [Hotamisligil, 2006], tightening the adaptive medium from both directions.

3.3 Oncologic system: C bypass as the CDFT inverse

Cancer constitutes the adaptive medium bypass failure mode of CDFT. Malignant cells engineer themselves to lower C for proliferative processes: the Warburg effect [Warburg, 1956, Vander Heiden et al., 2009] lowers C_E by switching to glycolysis; NRF2 activation lowers C_R [Schieber and Chandel,

2014]; epigenetic deregulation lowers C_M for transcription [Cairns et al., 2011]. In CDFT terms: cancer cells lower C , raise D_{eff} , and operate near $\Phi \approx 1$ (near the ceiling of what their modified C architecture permits). This creates disproportionate vulnerability to multi-constraint perturbation: near-ceiling operation means there is no C reserve to absorb additional constraint increases.

3.4 Immunologic system: T-cell exhaustion as fast C upregulation

Immune activation drives T-cells toward aerobic glycolysis — a configuration where C_E and C_R are near their maximum. Under chronic antigen stimulation, J_{req} persistently exceeds $J_{\text{eff}} = \nabla X / (C_E + C_R)$. T-cell exhaustion features — PD-1, LAG-3, TIM-3 upregulation; reduced cytokine production; impaired proliferation — are the molecular implementation of AFL fast dynamics: the immune medium raises C (lowers D_{eff}) to protect against energetic and redox catastrophe. Exhaustion-associated epigenetic marks [Bird, 2007, Kaelin and McKnight, 2013] represent slow dynamics ($dC_i/dt = -\lambda_i J + R_i$) progressively eroding the capacity to reverse this C elevation. Figure 2 provides detailed analysis.

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Figure 2: T-Cell Exhaustion as AFL Adaptive C Upregulation. (A) Effector flux: acute response (healthy, C low) vs. chronic activation (exhaustion, C rising). Declining J_{max} (red dashed) reflects progressive C_E, C_R elevation; shaded = dissociation gap where $J_{\text{req}} > J_{\text{max}}$. PD-1/LAG-3/TIM-3 = molecular fast-dynamics C elevation mechanism. (B) Immunometabolic trajectory through C state space: Naïve ($C \ll 1$) → Effector ($C \rightarrow 0.7$) → Chronically activated ($C_R \rightarrow \text{ceiling}$) → Exhausted ($C \gg J_{\text{req}}$, $D_{\text{eff}} \approx 0$). **KEY:** $J_{\text{eff}} = \nabla X_{\text{immune}} / (C_E + C_R)$; exhaustion = adaptive C upregulation protecting energetic and redox structural integrity.

3.5 Neurological system: dual- C firing rate saturation

Neural firing rate is subject to two CDFT/AFL constraint components:

$$J_{\text{neural}} \leq \frac{\nabla X_{\text{neural}}}{C_T + C_E} \quad (5)$$

C_T governs ion channel refractory period (transport constraint), defining maximum firing frequency independently of stimulus intensity. C_E governs ATP required for Na/K ATPase restoration of ion gradients — one of the highest energy-density biological processes [Attwell and Laughlin, 2001, Raichle and Gusnard, 2002]. The neural sigmoid activation function [Dayan and Abbott, 2001] is a CDFT special case: $f(I) = 1/(1 + e^{-I})$ is the saturation equation under simultaneous C_T and C_E constraints. The Shannon–Hartley information-theoretic limit [Shannon, 1948] is the CDFT ceiling when both C components are simultaneously at maximum.

3.6 Endocrine system: GH resistance as C_E dominance

GH drives anabolic processes requiring substantial energetic capacity. During fasting, C_E rises as metabolic fuel availability decreases. GH signalling raises $\nabla X_{\text{anabolic}}$ (the hormonal driving gradient) but the adaptive medium has simultaneously elevated C_E , such that $dC_E/dI_{\text{GH}} = 0$:

$$J_{\text{anabolic}} \leq \frac{\nabla X_{\text{anabolic}}}{C_E} \quad (6)$$

The hypothalamic-pituitary axis responds to low IGF-1 output by increasing ∇X_{GH} (raising GH), without recognizing that C_E is the binding constraint [Moller and Jorgensen, 2009]. In CDFT language: the system is escalating the driving gradient in a medium whose C is fixed by energetic state, not by the hormone.

4 Cross-System Comparison

Table 1 summarizes the CDFT/AFL architecture across all twelve biological systems.

Table 1: **Cross-System CDFT/AFL Architecture.** All twelve systems implement $F = \nabla X/C(\Omega, t)$ with domain-specific C architectures. In all systems, input-output dissociation occurs when the dominant C component is independent of the input signal being escalated.

System	Dominant C	Dissociation mechanism	CDFT/AFL equation
Haematologic	C_M, C_E, C_R	EPO raises ∇X but not C	$J_{\text{RBC}} \leq \nabla X / (C_M + C_E + C_R)$
Metabolic	C_E, C_R	Insulin raises ∇X ; $C_{E,R}$ fixed	$J_{\text{gluc}} \leq \nabla X / (C_E + C_R)$
Oncologic	All C lowered	C bypassed; D_{eff} raised	$C^{\text{cancer}} < C^{\text{normal}}; \Phi \approx 1$
Immunologic	C_E, C_R	Cytokines raise ∇X ; C fixed by state	Fast dynamics: $dC/dt = \alpha F - \beta C$
Neurological	C_T, C_E	Stimulus raises ∇X ; refractory/ C_E fixed	Dual: C_T (refractory) + C_E (ATP)
Endocrine	C_E	GH raises ∇X ; C_E fixed by energy state	$J_{\text{anab}} \leq \nabla X / C_E$
Microbiological	C_M, C_E (cost)	Resistance raises C ; moves when $U \rightarrow 0$	$C^{\text{resist}} = C_0 + C_{\text{resist}}$
Embryological	C_E, C_R, C_T, C_S	Instructions present; $C_{\text{dev}}(x, t)$ too high	$J_{\text{dev}} \leq \nabla X / C_{\text{dev}}(\Omega, x, t)$
Stem cell	C_E, E_p, B_e	Injury signal present; C_{regen} elevated	Slow drift: $dC_i/dt = -\lambda_i J + R_i$
Neural plasticity	C_E, C_T, E_p	Learning signal present; C_{plast} elevated	$P(\Omega) = 1/C_{\text{plast}} = g(E_n, B_e, E_p)$
Pharmacological	C_S, C_E, C_M	Dose raises ∇X ; C fixed by receptor state	Saturation + toxicity escalation
Ecological	C_{env}	Reproduction raises ∇X ; $C_{\text{env}} = f(K)$	$dN/dt = rN(1 - N/K); K = \nabla X / C_{\text{env}}$

5 Cross-System Predictions

Figure 3 illustrates the predicted patterns.

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Figure 3: **Cross-System CDFT/AFL Predictions.** **(P1)** C_i co-elevation (high Φ_i) across domains in multi-system disease. **(P2)** Sequential C relief staircase: each step reduces one dominant C component, producing a new haemoglobin plateau. **(P3)** C -profile PCA: patients cluster by dominant C mechanism ($C_E + C_R$ vs. C_M vs. C_T dominant), not by organ system. **KEY:** All predictions are CDFT/AFL predictions: measure C , not ∇X ; reduce C , not escalate ∇X .

P1: Cross-system C co-elevation in multi-system disease. Patients with multiple simultaneous resistance phenotypes will demonstrate co-elevated C -proxy measurements across multiple domains relative to single-system patients. In CDFT terms: when the adaptive medium is globally stiffened (elevated C throughout), all systems show simultaneous D_{eff} reduction. *Test:* CKD cohort comparing composite C_{Ω} between single-system and multi-system dysfunction. *Falsification:* No C_{Ω} co-elevation after confound adjustment.

P2: Sequential C relief staircase. Multi- C therapy will produce a stepwise staircase: each intervention that reduces one C component (increasing D_{eff} in that domain) produces a new haemoglobin plateau until the next-dominant C becomes rate-limiting. *Test:* Stepped trial: (1) iron/folate/B12 (reduce C_M); (2) add mitochondrial support (reduce C_E); (3) add anti-inflammatory (reduce C_R). Serial OCR, GSH/GSSG, SAM/SAH identify dominant C at each step. *Falsification:* Continuous linear response; no sequential plateaus.

P3: C -profile clustering is mechanism-based, not organ-based. Patients with different organ diagnoses but the same dominant C component will cluster together in C -profile space. In CDFT terms: disease classification should follow the C architecture of the adaptive medium, not the anatomical location of the symptom. *Test:* Multi-domain C_{Ω} profiling in CKD, metabolic syndrome, and autoimmune disease; unsupervised clustering should correlate with dominant C component more strongly than with conventional diagnosis. *Falsification:* Clusters align with conventional classification.

P4: Capacity-expansion (multi- C reduction) outperforms input escalation in multi-system disease. Interventions that simultaneously reduce multiple C components (C_E through mitochondrial biogenesis, C_R through antioxidant restoration, C_M through methylation optimization) will produce simultaneous improvement across multiple resistance phenotypes; input-escalation strategies that raise ∇X without reducing C will not. *Test:* In patients with co-occurring ESA hyporesponsiveness and insulin resistance (chronic C_E upregulation), compare input-escalation vs. capacity-expansion therapy; measure haemoglobin and glucose disposal simultaneously. *Falsification:* Input-escalation produces equivalent multi-system improvement.

P5: Maternal multi- C reduction decreases developmental anomaly risk beyond folate alone. In at-risk pregnancies, the developmental adaptive medium has globally elevated C (elevated C_E , C_M , C_R). Folate supplementation reduces C_M only. Comprehensive multi- C reduction — targeting C_E , C_M , and C_R simultaneously — will produce greater NTD and structural anomaly reduction by restoring D_{eff} across the full embryonic adaptive medium. *Test:* Randomized trial in metabolically at-risk pregnancies; primary outcome: structural anomaly rate at 20-week morphology scan. *Falsification:* Multi- C optimization produces no additional benefit beyond standard folate supplementation.

6 Main Results

6.1 Domain dictionary

System	Driver / φ proxy	Throughput Φ	Constraint C
Metabolism	chemical potential	substrate / ATP flux	enzyme / mitochondrial cap
Neural	membrane potential	ion current / spike budget	channel / fatigue load
Circulatory	pressure head	blood flow	vascular resistance
Immune	cytokine field	signaling flux	regulatory / inflammatory load
Development	morphogen potential	effective diffusion flux	receptor / GRN limits
Ecology	resource potential	energy / biomass flux	environmental / trophic cap

Status: Inferred

6.2 Cross-system predictions

If B1–B3 hold approximately, then: (i) saturation under increased drive; (ii) bottlenecking by the tightest effective constraint; (iii) adaptive adjustment of C following (2); (iv) operating bands expressed via $\Psi_s = (\Phi/C) \cdot S \cdot M_s$ in consistent units.

Status: Inferred

7 Numerical Verification

An active-numbered public supplement (NumPy, seed 7) instantiates four domains with identical lattice dynamics and distinct (α, β) , source strength, and boundary placement. Console Table 1 reports final Ψ_s , mean Φ , mean C , and a discrete regime label. Table 2 states the isomorphism claim in prose tied to those numbers.

Status: Derived

8 Interpretation

The result is **structural**: we do not claim that a single numeric Ψ_s is measurable or comparable across organs without calibration. Rather, the same *equations* reproduce homeostasis ($\Psi_s \approx 1.0$)-like bands when parameters are tuned per domain—consistent with a transferable modeling habit rather than a new empirical law.

Status: Inferred

9 Limitations and Open Problems

- No statistical fit to experimental time series.
- Discrete regime thresholds in the script are illustrative.
- Regulatory networks with sharp switches may need hybrid models.

Status: Open

10 Conclusion

AFL-2 extends AFL-1 by exhibiting a shared AFL closure across stylized biological domains. The next paper (AFL-3) anchors the same contract in a clinical pathway: erythropoiesis in chronic disease.

Status: Inferred

11 Reproducibility Note

- **Script packaging:** active-numbered public supplement.
- **Dependencies:** NumPy; runtime $\ll$ 1 s.
- **Seed:** 7.
- **Outputs:** console Tables 1–2 (final Ψ_s , means, regime labels).

Status: Derived

12 Deepening the Universality Claim

AFL should not be interpreted as a claim that every biological process is metabolic in disguise. The stronger claim is structural: many biological systems maintain function by matching throughput to adaptive constraint. A neural circuit, an immune response, a vascular bed, a developmental field, and an ecosystem differ materially, but each can enter overload when drive exceeds regulatory capacity.

The universal biology claim therefore requires three tests. First, the system must have a measurable throughput variable. Second, it must have a measurable constraint or capacity variable. Third, perturbing throughput must change the constraint state, either through adaptation, fatigue, repair, suppression, or remodeling. If any of these fail, AFL should not be applied.

13 Cross-System Research Program

A strong biology program should compare domains by nondimensionalized trajectories rather than raw units. For each domain, one can measure a normalized flux $\widehat{\Phi}$, a normalized constraint $\widehat{C}$, and the operating ratio $\widehat{\Psi}_s = \widehat{\Phi}/\widehat{C}$. The prediction is not that the numbers match across organs; it is that the phase portrait has recurring regions: under-drive, regulated throughput, overload, compensatory constraint rise, and decompensation.

This gives a practical research design. In metabolism, measure substrate turnover, redox burden, and mitochondrial reserve. In immunity, measure antigenic or inflammatory drive against regulatory and tissue-tolerance capacity. In development, measure morphogen throughput against receptor, transcriptional, and mechanical constraints. In ecology, measure energy transfer against trophic and environmental limits. If these systems produce comparable phase portraits after normalization, AFL becomes a biological systems principle rather than a metabolic analogy.

14 Falsifiability

The universality claim fails where throughput and constraint cannot be operationalized, where constraint does not respond to flux, or where observed transitions are better explained by fixed thresholds without adaptation. The framework also fails if it cannot predict the direction of regime change under intervention. A useful AFL model must say whether reducing flux, lowering constraint, increasing relaxation, or redistributing load should improve the system.

15 Clinical Mapping of Symbols

Symbol	Universal meaning	mean-	Candidate (domain-dependent)	proxy	Invalidation condition
Φ	Throughput	in-	Substrate flux, signal-		If Φ cannot be operational-
	intensity		ing rate, energy trans-		ized in a given domain,
C	Constraint / ca-		Enzyme capacity,		AFL should not be applied
	capacity burden		vascular resistance,		If C does not respond to
			trophic limit		Φ perturbation, no adap-
					tation — static model suf-
					fices
S	Responsiveness		Normalized adapta-		If constant across domains,
			tion index; domain-		not useful for cross-domain
			specific		comparison
M_s	Memory / locked		Domain-specific: fi-		If $M_s \rightarrow 0$ rapidly, memory
	state		bro-sis, epigenetic age,		framing is not needed
			scar tissue		
Ψ_s	Operating state		Computed proxy; not		If phase portrait does not
			a single universal assay		match AFL prediction af-
					ter normalization, frame-
					work fails for that domain

Status: Hypothesis

16 Clinical Safety Boundary

AFL-2 is a research framework establishing structural universality of the flux-constraint grammar. It does not provide cross-domain clinical decision thresholds. Ψ_s values are not comparable across organ systems without domain-specific calibration. No treatment decisions should be based on AFL framing without domain-specific clinical validation.

17 Runtime Diagnostic

Release-local script: `supplementary/supplementary_afl_02.py`. Outputs in `outputs/paper_02/:` `domain_comparison.csv`, `cross_domain_afl.png`, `summary.json`. All outputs are toy diagnostics showing candidate cross-domain AFL mappings — not validated measurements.

Conflict of Interest

The author declares no conflict of interest.

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Author Contributions

Conceptualization, formal analysis, runtime engineering, and manuscript preparation: S.B.M.

Data Availability

All computational models and scripts are in the `supplementary/` directory.

18 Reference Layer

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